

# Chapter:5

## DC Machine

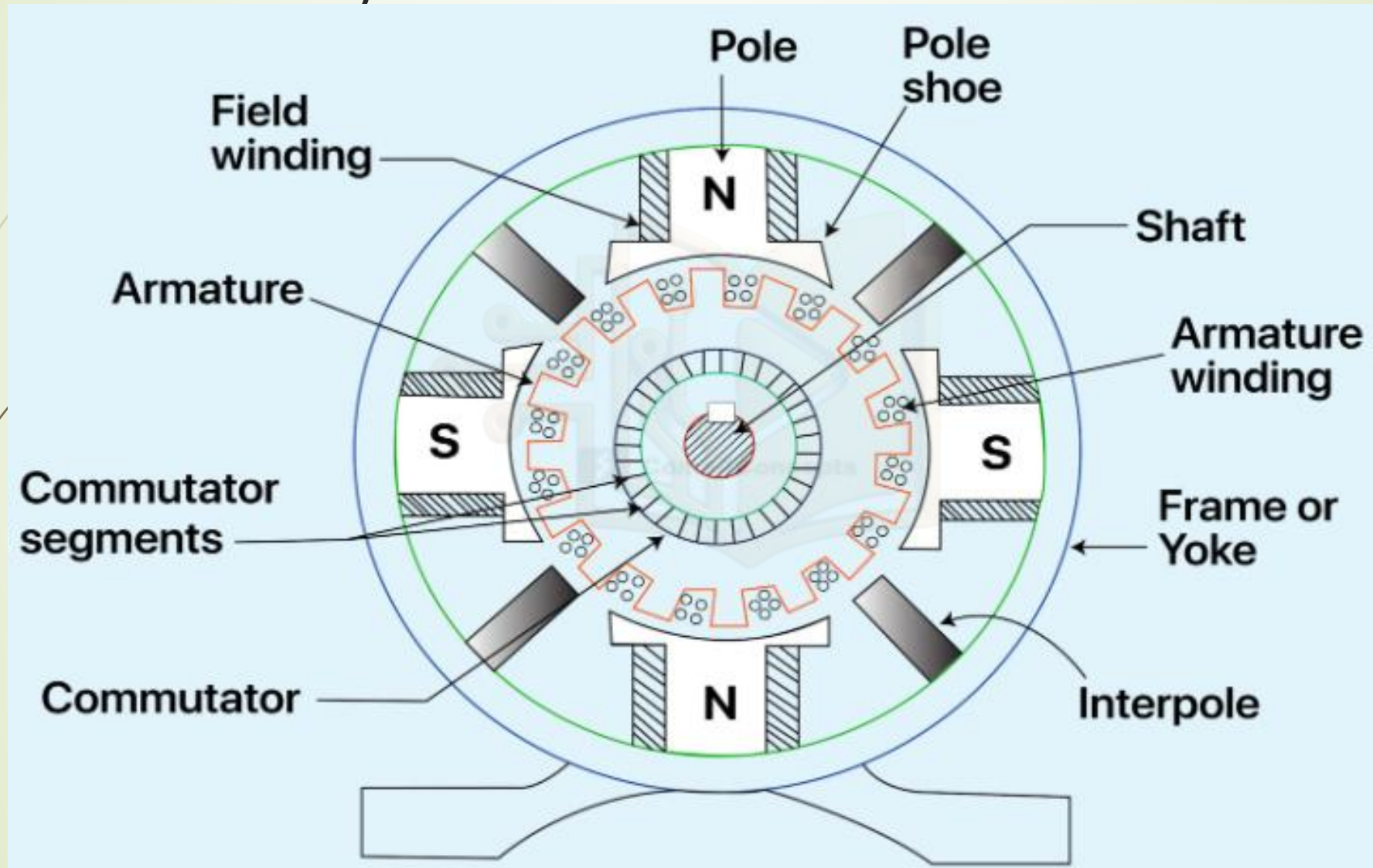
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## Introduction

- DC machine is an **electromechanical energy alteration device**
- DC machine that converts mechanical power to electrical power are **Generators**, where as a DC machine that converts electrical power to mechanical power are **Motors**
- DC machine is most frequently used for a motor
- The main benefits of this machine include torque regulation as well as easy speed regulation.
- The **applications of the DC machine** is limited to trains, mills, and mines
- For example, underground subway cars, as well as trolleys, may utilize DC motors. In the past, automobiles were designed with DC dynamos for charging their batteries

# Construction of DC machines (Generator and Motor)



## 1. Yoke (Magnetic Frame)

- The outer frame or yoke serves a double purpose
- It provides **mechanical support** for the poles and acts as a protecting cover for the whole machine
- It **carries the magnetic flux** produced by the poles
- yokes are made of **cast iron** or **steel**

## 2. Field system(Stator)

- Stator is the **stationary part** of the machine
- Its main function is to **provide main magnetic flux** in which **armature coil rotates**.
- It consists of poles and pole shoes

### Pole core and Pole Shoes

- The field magnets consist of **poles (north south poles of magnet) and pole shoes**.
- The pole shoes serve two purposes:
  - I. they **spread out the flux** in the air gap
  - II. they support the **exciting coils (or field coils)**



## Field windings

- It is the winding that produces **main flux** in the machine.
- Field winding is applied to **stator**.
- These windings are **wound on poles** in such a way that whenever current passes through it they act like **electromagnet**.

## 2. Armature System(Rotor)

- Rotor is the **basic part** of DC machine.
- It consists of armature core, armature winding and commutator.

### Armature

- It is the **rotating part** of dc machine.
- It is **cylindrical** in shape.
- It consists of **core that is laminated** to reduce eddy current loss.
- **Core slots** are also present that **holds armature windings**

## Armature windings

- These are the conductor **winding placed in the slots** provided in the armature core.
- When armature core rotates, **EMF is induced** in the armature winding.
- **Coil ends** of armature windings are connected to **commutator segments**.

### 3. Commutator

- It is a **mechanical rectifier** used for obtaining a rectified waveform at the output in order to get dc voltage.
- It is responsible to **supply current** to the **load** through brushes.
- The commutator is made of **copper segments** insulated from each other by mica sheets and mounted on the shaft of the machine.

### 4. Brushes

- It helps to **gather the current from the commutator and supply it to the exterior load**.
- **It acts as a contact**.

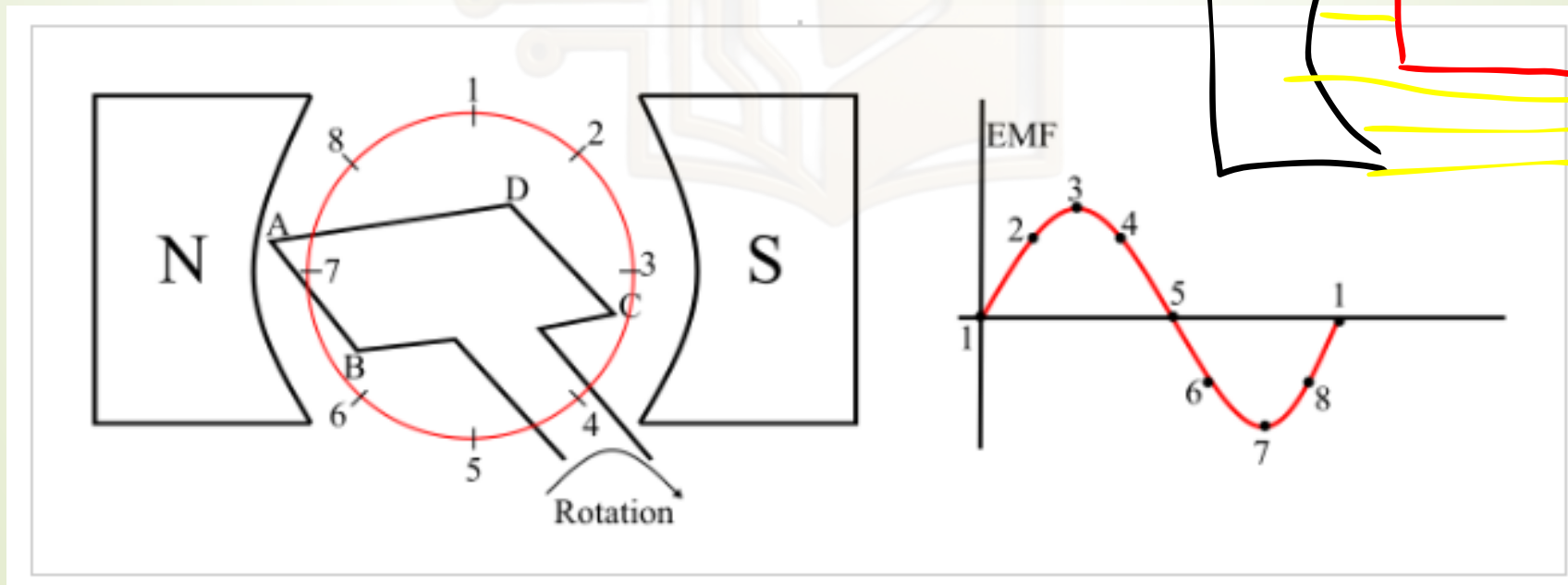
# DC Generator

- It is an **electromechanical energy** conversion device that converts **mechanical power into DC electrical power** through the process of electromagnetic induction.
- A DC generator operates on the **principle of electromagnetic induction**
- When the magnetic flux linking a conductor changes, an EMF is induced in the conductor.
- A DC generator has a **field winding and an armature winding**.
- The **armature winding** of DC Generator is placed on the **rotor** whereas the **field winding** is placed on the **stator**.
- **EMF induced** in the armature winding of a **DC generator is alternating** and is converted into **direct voltage using a commutator**

# Working principle of DC generator

Consider a single loop DC generator (as shown in the figure), in this a single turn loop 'ABCD' is **rotating clockwise** in a uniform magnetic field with a constant speed.

- When the loop rotates, the magnetic flux linking the coil sides 'AB' and 'CD' changes continuously.
- This **change in flux linkage** induces an **EMF in coil** sides and the induced EMF in one coil side adds the induced EMF in the other.





The EMF induced in a DC generator can be explained as follows

- When the loop is in position-1, **the generated EMF is zero** because, the movement of coil sides is parallel to the magnetic flux.
- When the loop is in position-2, the coil sides are moving at an angle to the magnetic flux and hence, a **small EMF is generated**.
- When the loop is in position-3, the coil sides are moving at **right angle** to the magnetic flux, therefore the **generated EMF is maximum**.
- When the loop is in position-4, the coil sides are cutting the magnetic flux at an angle, thus a **reduced EMF** is generated in the coil sides.
- When the loop is in position-5, no flux linkage with the coil side and are **moving parallel to the magnetic flux**. Therefore, no EMF is generated in the coil.
- At the position-6, the coil sides move under a pole of opposite polarity and hence the **polarity of generated EMF is reversed**. The maximum EMF will generate in this direction at position-7 and zero when at position-1.
- This cycle repeats with revolution of the coil.

- It is clear that the generated EMF in the loop is **alternating one**.
- It is because any coil side (say AB) has EMF in one direction when under the influence of N-pole and in the other direction when under the influence of S-pole.
- Hence, when **a load** is connected across the terminals of the generator, **an alternating current will flow through it**.
- And by using a commutator, this **alternating emf generated in the loop** can be converted into direct voltage

# EMF Equation

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- Let us suppose there are Z total numbers of conductor in a generator, and arranged in such a manner that all parallel paths are always in series.

Here,

Z = total numbers of conductor

A = number of parallel paths

Then,

Z/A = number of conductors connected in series

We know that induced emf in each path is same across the line

Therefore,

Induced emf of DC generator

$e = \text{emf of one conductor} \times \text{number of conductor connected in series.}$

- A=P for lap type of winding, A= 2 for wave type winding, p= number of poles
- Induced emf of DC generator is,

$$e = \phi P \frac{N}{60} \times \frac{Z}{A} \text{ volts}$$

# Types of DC Generator

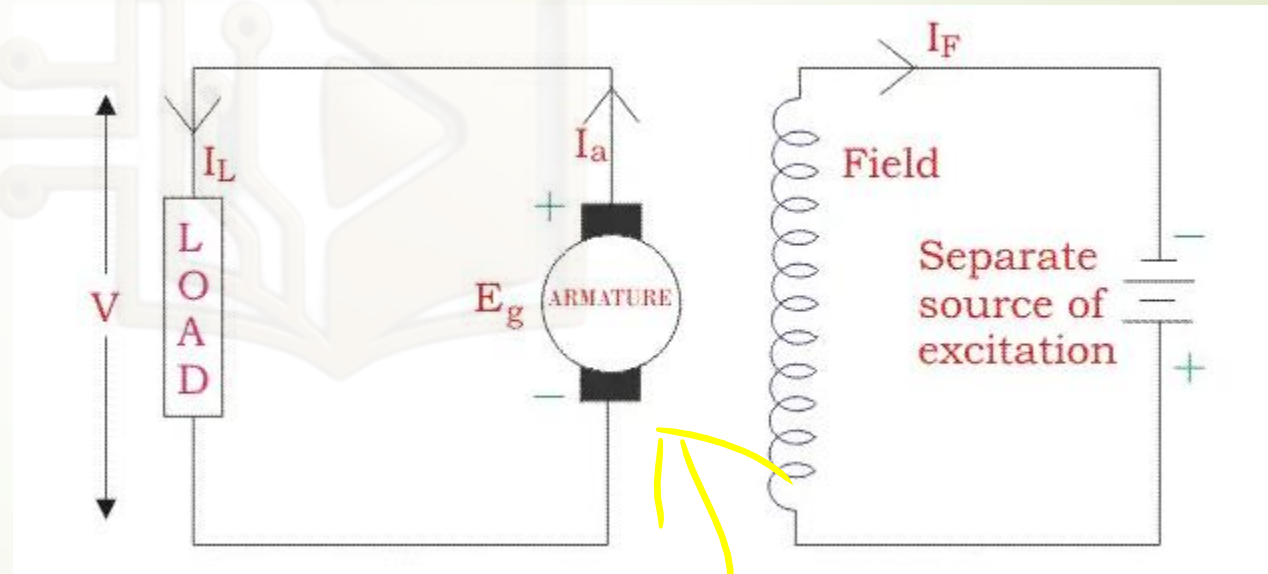
- DC generators are classified based on how their fields are excited (i.e. produced). There are three methods of excitation, and thus three main types of DC generators:
- Permanent Magnet DC Generators : **Field coils excited by permanent magnets**
- Separately Excited DC Generators : **Field coils excited by some external source**
- Self Excited DC Generators: **Field coils excited by the generator itself**

Self-excited DC generators can further be classified depending on the position of their field coils. The three types of self-excited DC generators are:

- Series Wound Generators
- Shunt Wound Generators
- Compound Wound Generators

# Separately Excited DC Generator

- These are the generators whose field magnets are energized by some external DC source, such as a battery.
- A circuit diagram of separately excited DC generator is shown in the figure below. The symbols below are:
  - $I_a$  = Armature current
  - $I_L$  = Load current
  - $V$  = Terminal voltage
  - $E_g$  = Generated EMF



$$E_g = I_a R_a + V$$

$R_a$



➤ Voltage drop in armature =  $I_a \cdot R_a$

➤ Let,

$$I_a = I_L = I \text{ (say)}$$

➤ Then,

~~voltage across the load,  $V = E_g - I R_a$~~

➤ Power Generated is equal to,

$$P_g = E_g \times I$$

➤ Power delivered to external load is equal to,

$$P_L = V \times I$$

# Self Excited DC Generators

## ➤ Series Wound Generator

➤ In these type of generators, **the field windings are connected in series with armature** conductors, as shown in the figure below.

➤ Whole current flows through the field coils as well as the load. As series field winding carries full load current it is designed with relatively few turns of thick wire.

➤ Here:

➤  $R_{sc}$  = Series winding resistance

➤  $I_{sc}$  = Current flowing through the series field

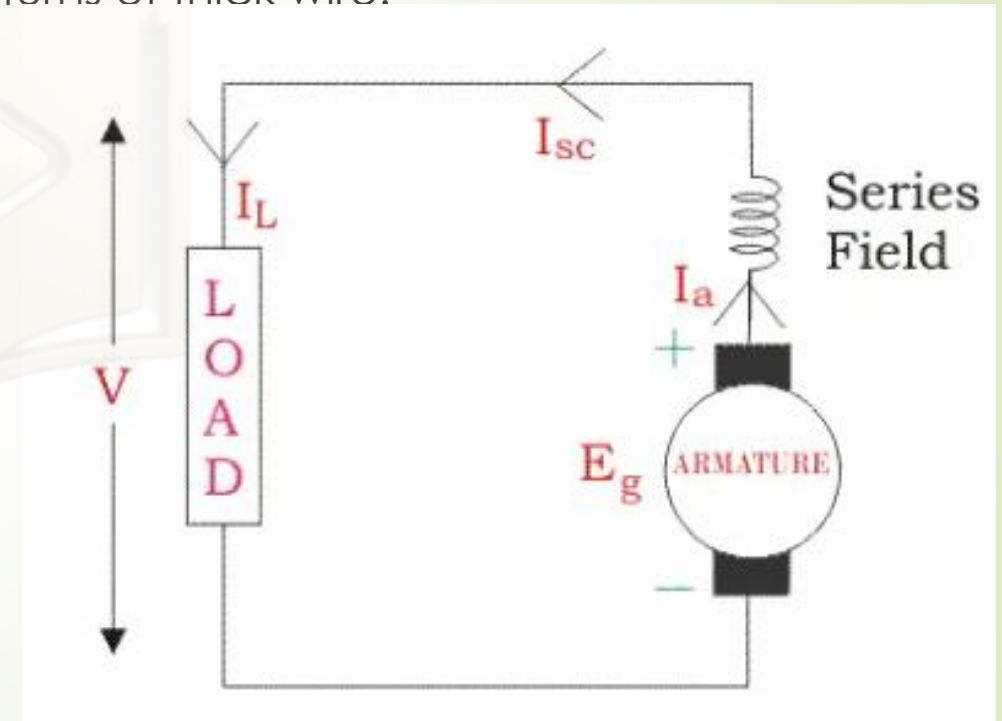
➤  $R_a$  = Armature resistance

➤  $I_a$  = Armature current

➤  $I_L$  = Load current

➤  $V$  = Terminal voltage

➤  $E_g$  = Generated EMF



- Then,

$$I_a = I_{sc} = I_L = I \text{ (say)}$$

- Voltage across the Load is equal to,

$$V = E_G - I(R_a + R_{se})$$

- Power generated is equal to ,

$$P_g = E_g \times I$$

- Power delivered to load,

$$P_L = V \times I$$

$$E_g = I_a(R_a + R_s) + V$$

## ➤ Shunt Wound DC Generators

➤ In these type of DC generators, the **field windings are connected in parallel with armature conductors**, as shown in the figure below. In shunt wound generators the voltage in the field winding is same as the voltage across the terminal.

➤ Here:

➤  $R_{sh}$  = Shunt winding resistance

➤  $I_{sh}$  = Current flowing through the shunt field

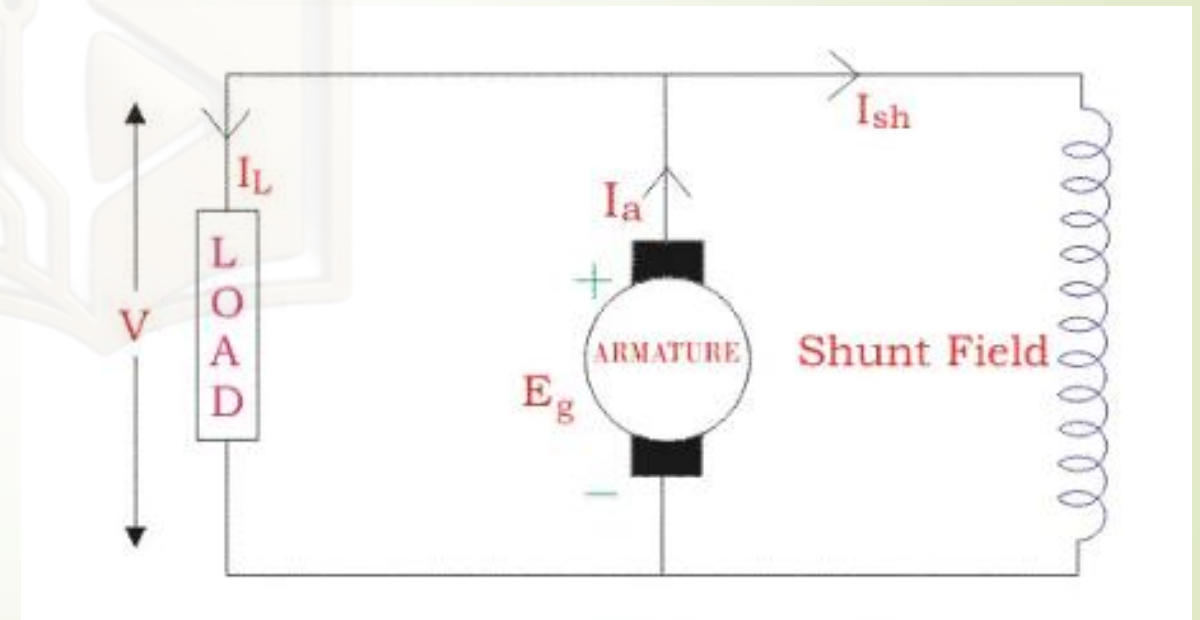
➤  $R_a$  = Armature resistance

➤  $I_a$  = Armature current

➤  $I_L$  = Load current

➤  $V$  = Terminal voltage

➤  $E_g$  = Generated EMF



- Here armature current  $I_a$  is dividing in two parts – one is shunt field current  $I_{sh}$ , and another is load current  $I_L$ .

- So,

$$I_a = I_{sh} + I_L$$

- Shunt field current,

$$I_{sh} = \frac{V}{R_{sh}}$$

- Voltage across the load

$$V = E_g - I_a R_a$$

- Power generated is

$$P_g = E_g \times I_a$$

- Power delivered to the load is,

$$P_L = V \times I_L$$

$$I_a = I_{sh} + I_L$$

$$E_g - I_a R_a - V = 0$$

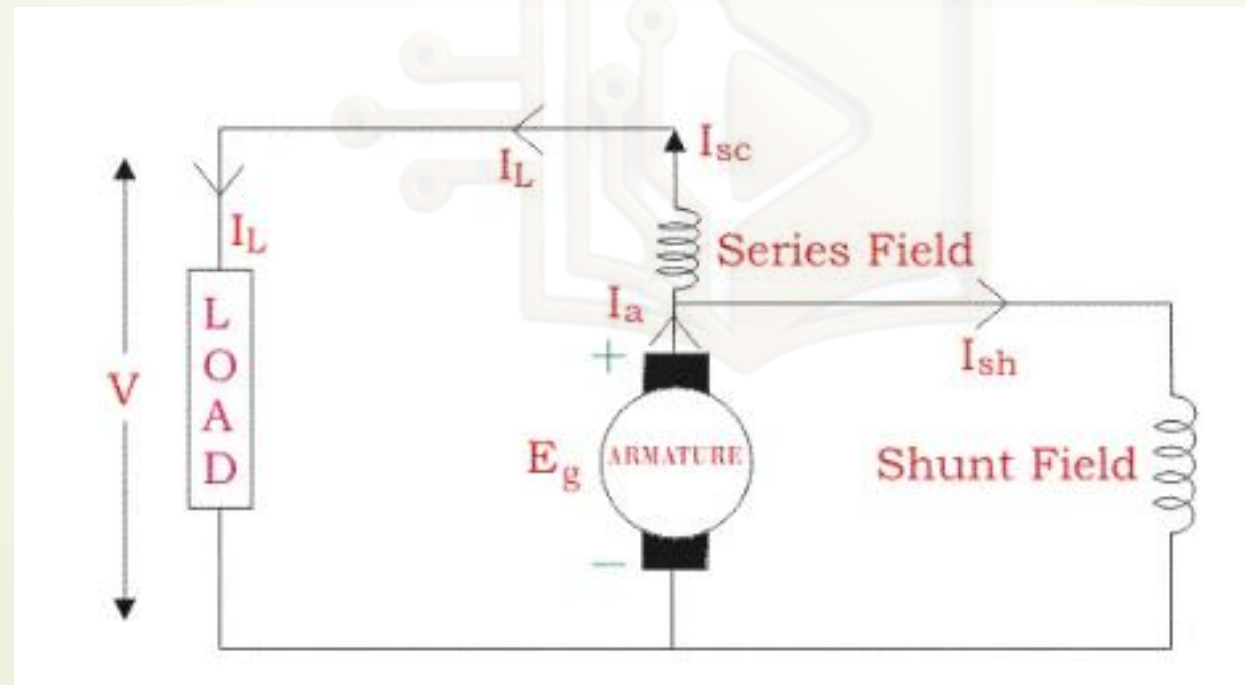


## ➤ Compound Wound DC Generator

- This type of DC generators may be of two types- short shunt compound-wound generator and long shunt compound-wound generator.

### Short Shunt Compound Wound DC Generator

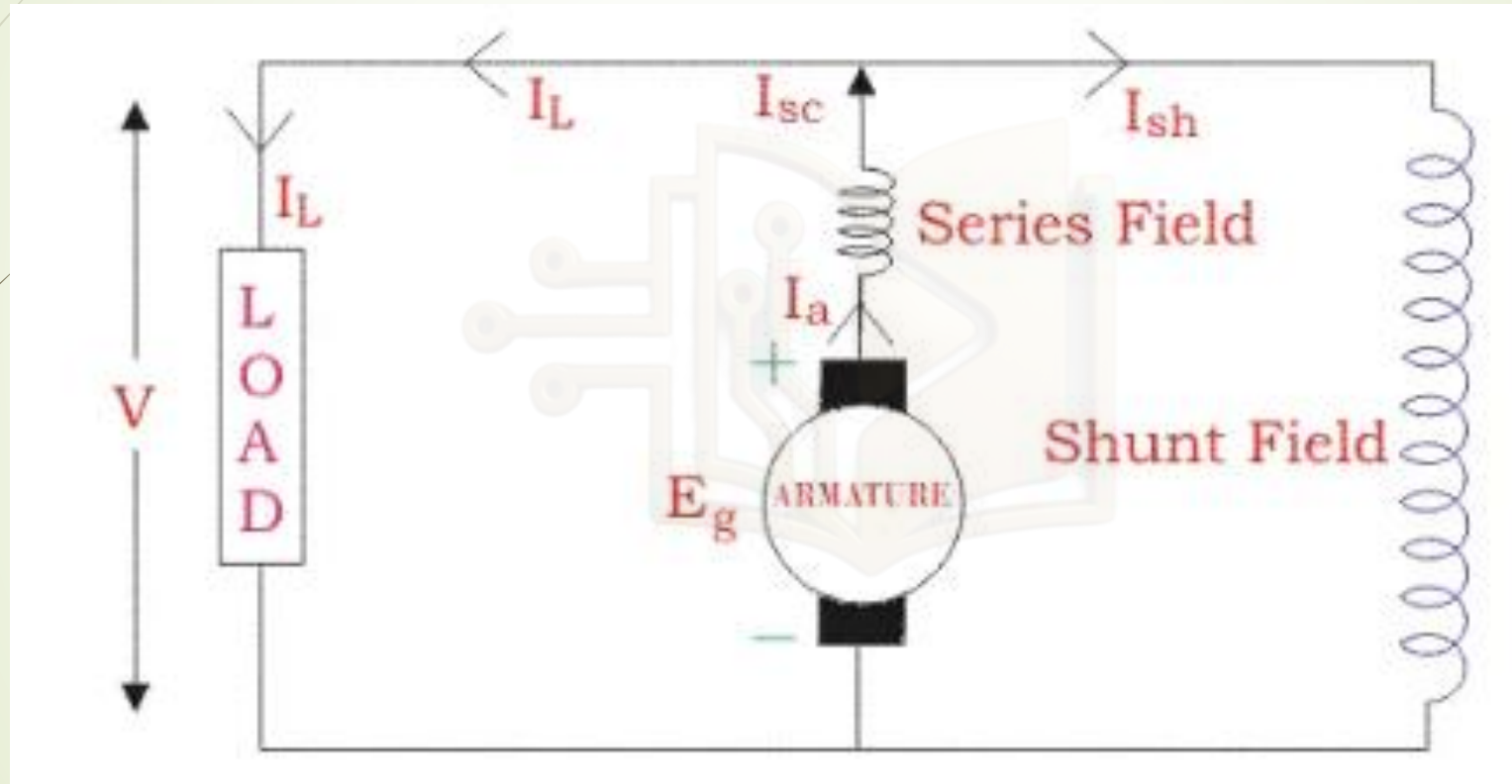
- Short Shunt Compound Wound DC Generators are generators **where only the shunt field winding is in parallel with the armature winding**, as shown in the figure below.



- Series Field current is,  $I_{sc} = I_L$
- Shunt field current is,  $I_{sh} = \frac{(V + I_{sc}R_{sc})}{R_{sh}}$
- Armature current is,  $I_a = I_{sh} + I_L$
- Voltage across the load is,  $V = E_g - I_a R_a - I_{sc} R_{sc}$
- Power generated is,  $P_g = E_g \times I_a$
- Power delivered to load is,  $P_L = V \times I_L$

## Long Shunt Compound Wound DC Generator

- Long Shunt Compound Wound DC Generator are generators where the **shunt field winding is in parallel with both series field and armature winding**, as shown in the figure below.



- Shunt field current,

$$I_{sh} = \frac{V}{R_{sh}}$$

- Armature current,

$$I_{sc} = I_L + I_{sh}$$

- Voltage across the load is,

$$V = E_g - I_a R_a - I_{sc} R_{sc} = E_g - I_a (R_a + R_{sc}) [\because I_a = I_{sc}]$$

- Power generated is,

$$P_g = E_g \times I_a$$

- Power delivered is,

$$P_L = V \times I_L$$

$$E_g - I_a R_a - I_{sh} R_{sh} - V = 0$$

$$I_{sh} = \frac{V}{R_{sh}}$$

# DC motor and its working principle

- A DC motor is an **electromechanical energy conversion device**, which converts **electrical energy input into the mechanical energy** output.
- Consider a two pole DC motor. When the DC motor is connected to an external source of DC supply, the field coils are excited developing alternate N and S poles and a current flows through the armature windings.
- All the armature conductors under **N pole carry current in one direction** whereas all the conductors under **S pole carry current in the opposite direction**.
- As each **conductor carrying a current is placed in a magnetic field**, hence a **mechanical force acts** on it.
- The **force on all the conductors add together to exert a torque** which make the armature rotating.



# Motor torque

- Torque is the **tendency of a force to rotate or move** an object about an axis.
- A force is a push or pull, likewise, torque is a twist to an object. Mathematically, torque,  $T = F \times r$ .
- Let  $T_g$  = armature or gross torque (N-m) = Force  $\times$  radius.
- $r$  = radius of the armature in m.
- $N$  = speed of the armature in rpm =  $N/60$  rps.

*Work done/revolution = force  $\times$  distance moved per revolution*

$$w.d. = F \times 2\pi r \text{ Nm}$$

$$w.d. / s = F \times 2\pi r \times \frac{N}{60} \text{ Nm}$$

$$w.d. = \frac{2\pi N}{60} (F \times r) \text{ Nm / s or watt}$$

$$w.d. = \frac{2\pi N T_g}{60} \text{ watt ... (1)}$$

- The expression for voltage in dc motor is given by,

$$V = E_a + I_a R_a$$

(multiply throughout by  $I_a$ )

$$V I_a = E_a I_a + I_a^2 R_a$$

- Electrical input = electrical power equivalent to mechanical power developed + armature copper loss
- Mechanical power developed,

$$= E_a I_a \text{ watt ... (2)}$$

- Since equation (1) = equation (2),

$$\frac{2\pi N T_g}{60} = E_b I_a$$

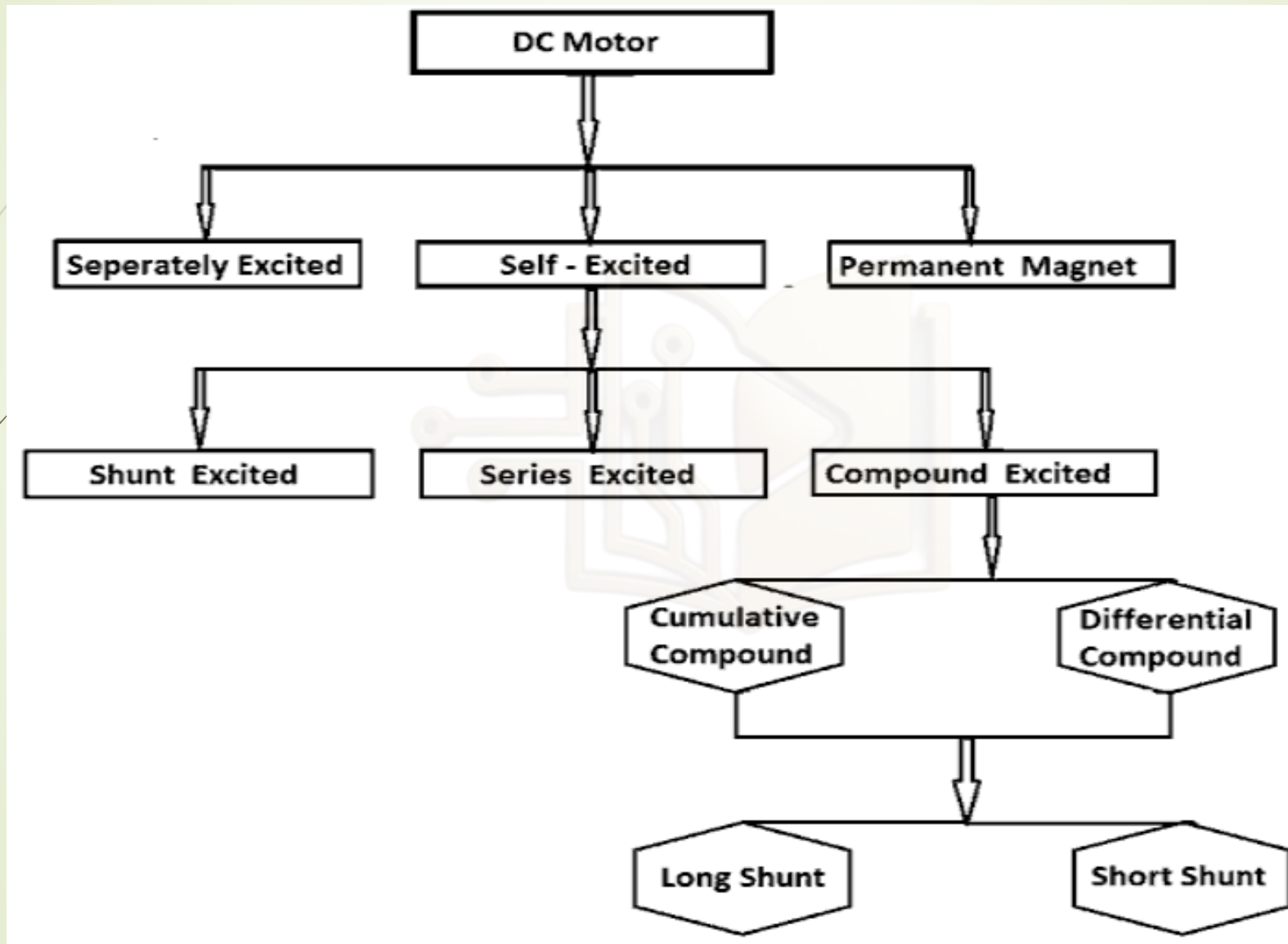
$$\frac{2\pi N T_g}{60} = \frac{\phi Z N}{60} \times \frac{P}{A} \times I_a$$

$$T_g = \frac{1}{2\pi} \phi Z I_a \times \frac{P}{A}$$

$$T_g = 0.159 \phi Z I_a \times \frac{P}{A} \text{ Nm}$$

# Types of DC motors

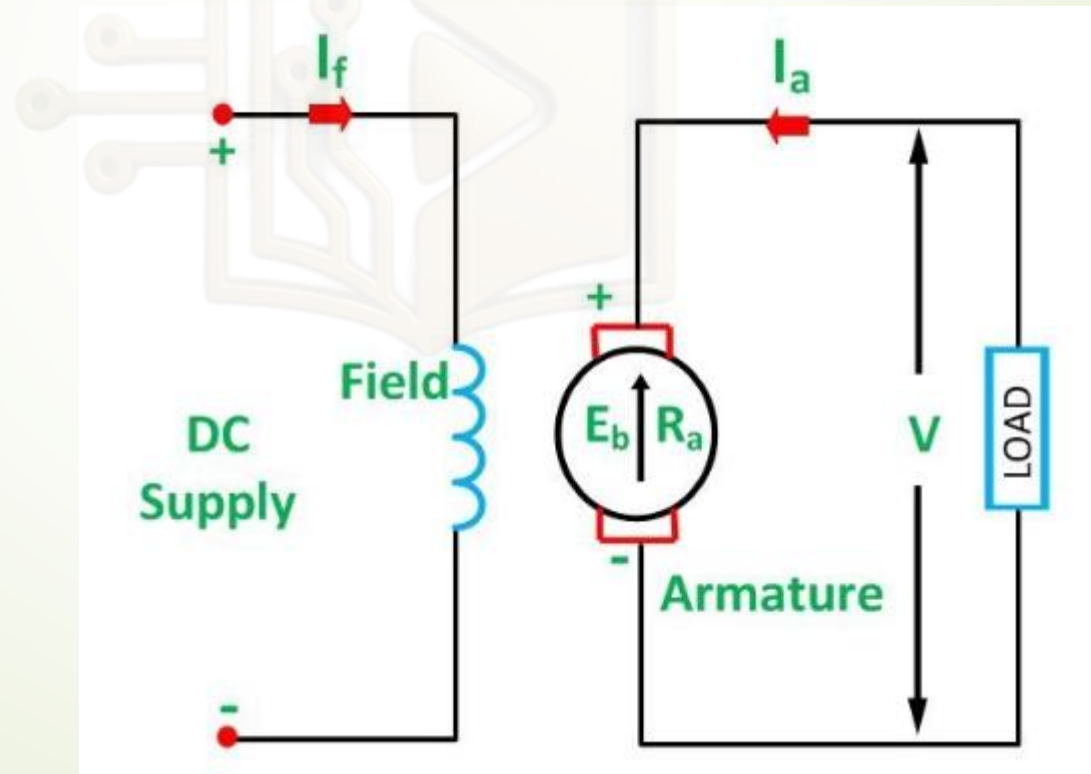
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## Separately Excited DC Motor

- The supply is given separately to the field windings.
- The armature current does not flow through the field windings, as the field winding is energized from a separate external source .
- We know  $T_g = K_a \phi I_a$
- So the torque in this case can be varied by varying field flux  $\phi$ , independent of the armature current  $I_a$ .



## Self Excited DC Motor

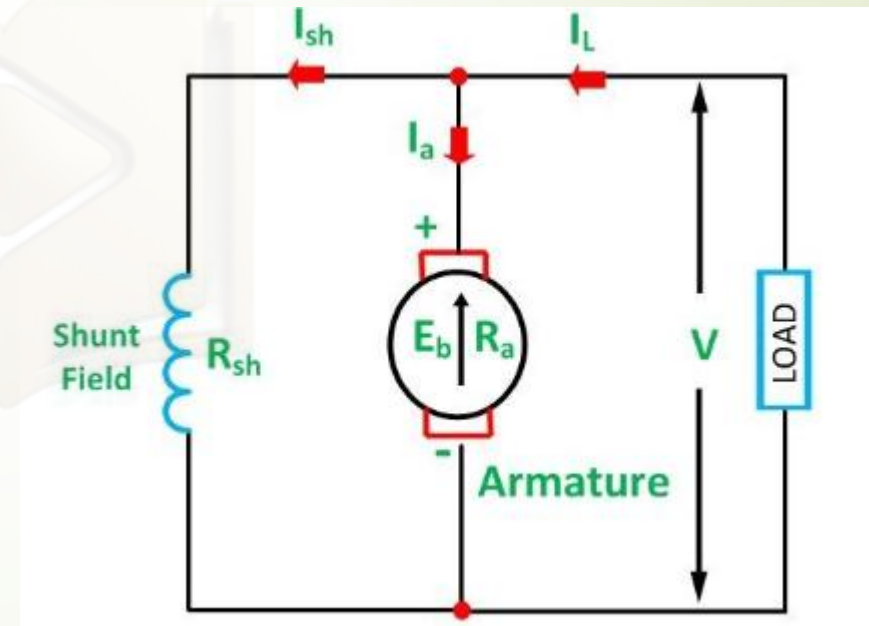
- In this type of motor, the current in the windings is supplied by the machine or motor itself.

### 1. Shunt Wound Motor

- This is the most common types of DC Motor. Here the field winding is connected in parallel with the armature.
- On applying KCL at the shown junction

$$I = I_a + I_{sh} \dots \dots \dots (1)$$

- Where,
- $I$  is the input line current
- $I_a$  is the armature current
- $I_{sh}$  is the shunt field current
- Equation (1) is the current equation.





The voltage equations are written by using Kirchhoff's voltage law (KVL) for the field winding circuit.

$$V = I_{sh}R_{sh} \dots \dots \dots (2)$$

For armature winding circuit the equation will be given as:

$$V = E + I_a R_a \dots \dots \dots (3)$$

The power equation is given as:

➤ **Power input = mechanical power developed + losses in the armature + loss in the field.**

$$VI = P_m + I_a^2 R_a + I_{sh}^2 R_{sh} \dots \dots \dots (4)$$

$$VI = P_m + I_a^2 R_a + VI_{sh}$$

$$P_m = VI - VI_{sh} - I_a^2 R_a = V(I - I_{sh}) - I_a^2 R_a$$

$$P_m = VI_a - I_a^2 R_a = (V - I_a R_a)I_a$$

$$P_m = EI_a \dots \dots \dots (5)$$

Multiplying equation (3) by  $I_a$  we get the following equations.

$$VI_a = EI_a + I_a^2 R_a \dots \dots (6)$$

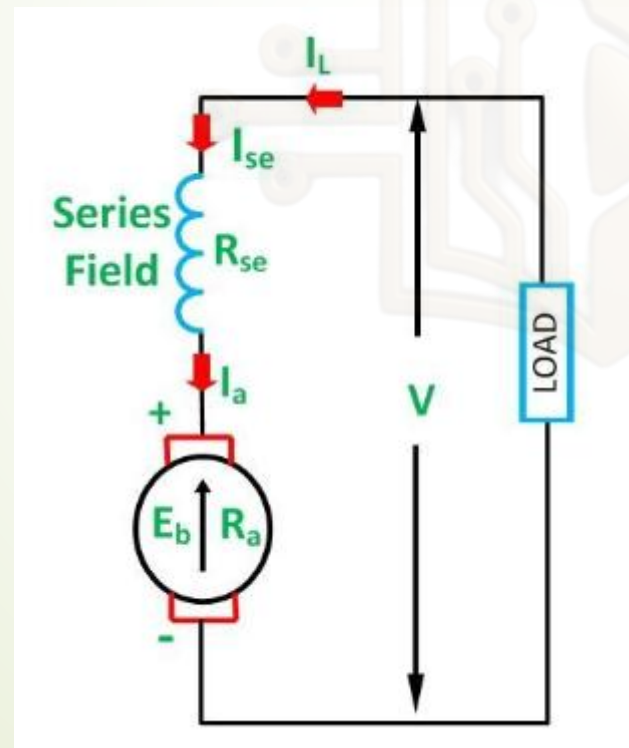
Where,

$$VI_a = P_m + I_a^2 R_a \dots \dots (7)$$

$VI_a$  is the electrical power supplied to the armature of the motor.

### Series Wound Motor

- In the series motor, the field winding is connected in series with the armature winding.



$E_b$   
back emf

By applying the KCL in the above figure:

$$I = I_{se} = I_a$$

Where,

➤  $I_{se}$  is the series field current

The voltage equation can be obtained by applying KVL in the above figure.

$$V = E + I(R_a + R_{se}) \dots \dots \dots (8)$$

The power equation is obtained by multiplying equation (8)

$$VI = EI + I^2(R_a + R_{se}) \dots \dots \dots (9)$$

Power input = mechanical power developed + losses in the armature + losses in the field

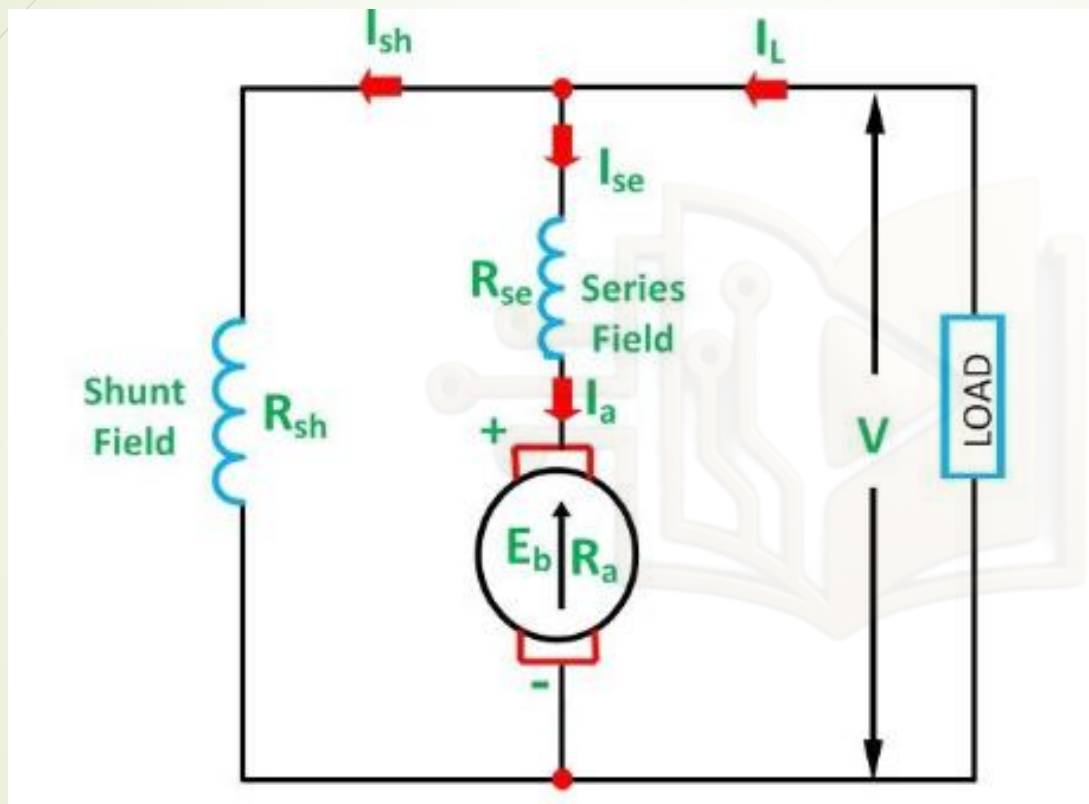
$$VI = P_m + I^2R_a + I^2R_{se} \dots \dots \dots (10)$$

Comparing the equation (9) and (10), we will get the equation shown below

$$P_m = EI \dots \dots \dots (11)$$

## Compound Wound Motor

A DC Motor having both shunt and series field windings is called a **Compound Motor**.



$$V - I_{se}R_{se} - I_aR_a - E_b = 0$$

# Speed Of A DC Motor (Theory)

- Back emf “e” of a DC motor is nothing but the induced emf in armature conductors due to rotation of the armature in magnetic field.
- Thus, the magnitude of  $E_b$  can be given by EMF equation of a DC generator.

$$e = \phi P \frac{N}{60} \times \frac{Z}{A} \text{ volts}$$

$E_b$  can also be given as,

$$E_b = V - I_a R_a$$

thus, from the above equations

$N = e \cdot 60 A / (P \phi Z)$  but, for a DC motor A, P and Z are constants

Therefore,  $N \propto K e / \phi$  (where, K=constant)

$$\begin{aligned} E &\propto \phi \cdot N \\ N &\propto E \\ N &\propto \frac{1}{\phi} \end{aligned}$$

This shows the **speed of a dc motor** is directly proportional to the back emf and inversely proportional to the flux per pole.



# Speed Control Of DC Motor

## Flux Control Method:

- As the **speed of a dc motor** is inversely proportional to the flux per pole.
- Thus by decreasing the flux, speed can be increased and vice versa.
- To control the flux, a rheostat is added in series with the field winding
- In case of **shunt motors adding more resistance in series** with the field winding will **increase the speed as it decreases the flux**.

## Armature Control Method

- **Speed of a dc motor** is **directly proportional to the back emf  $E_b$**  and  $E_b = V - I_a R_a$ .
- when supply voltage  $V$  and the armature resistance  $R_a$  are kept constant, then the speed is directly proportional to armature current  $I_a$ .
- Thus, if resistance is added in series with the armature,  $I_a$  decreases and, hence, the speed also decreases.
- Greater the resistance in series with the armature, greater the decrease in speed.